Class activity: regression interpretation and inference

Group members:

Instructions: Work with a neighbor on the following activity. I will collect the handout at the end of class, and it will be part of your class participation grade. You will be graded only on effort – it is ok if you don't finish all the questions, or get them all correct.

Breakfast cereal nutrition

In this activity, we have data on a sample of 36 different breakfast cereals. For each cereal, we have:

- `Sugar`: the grams of sugar per serving
- `Calories`: the calories per serving

Your parents, worried about the nutritional value of your breakfast, believe that there will be a positive relationship between the amount of sugar in a serving, and the number of calories per serving.

1. Write down a population linear regression model that would allow you to explore your parents' belief. You may assume that no transformations are needed on either variable.

You fit the model in R, producing the output below:

```
Coefficients:
            Estimate Std. Error
(Intercept)  87.4277     5.1627
Sugar         2.4808     0.7074
```

2. Interpret the estimated slope in context.

3. Write down null and alternative hypotheses, in terms of a parameter from your population model, that allow you to address your parents' question.

4. Calculate an appropriate test statistic for these hypotheses, using the R output shown above.

5. What will be the degrees of freedom for the t distribution used to calculate a p-value in this test?

6. Below is a picture of the appropriate t distribution. Shade the area under the curve corresponding to the p-value for your hypothesis test.